

Local AI on your OnePlus: three sizes of Qwen

AdHush can put one question, over and over, to a **language model that lives on the phone**: *is this a commercial or the show?* The phone listens to the TV, turns the speech into words, hands the last 40 seconds of them to the model, and receives a one-line answer whose confidence becomes a vote; a confident “commercial” turns the TV down. **Nothing leaves the phone, and it costs nothing.** This guide explains the three sizes of model the app can install, how to choose between them, and how to tell whether the one you chose is earning its keep.

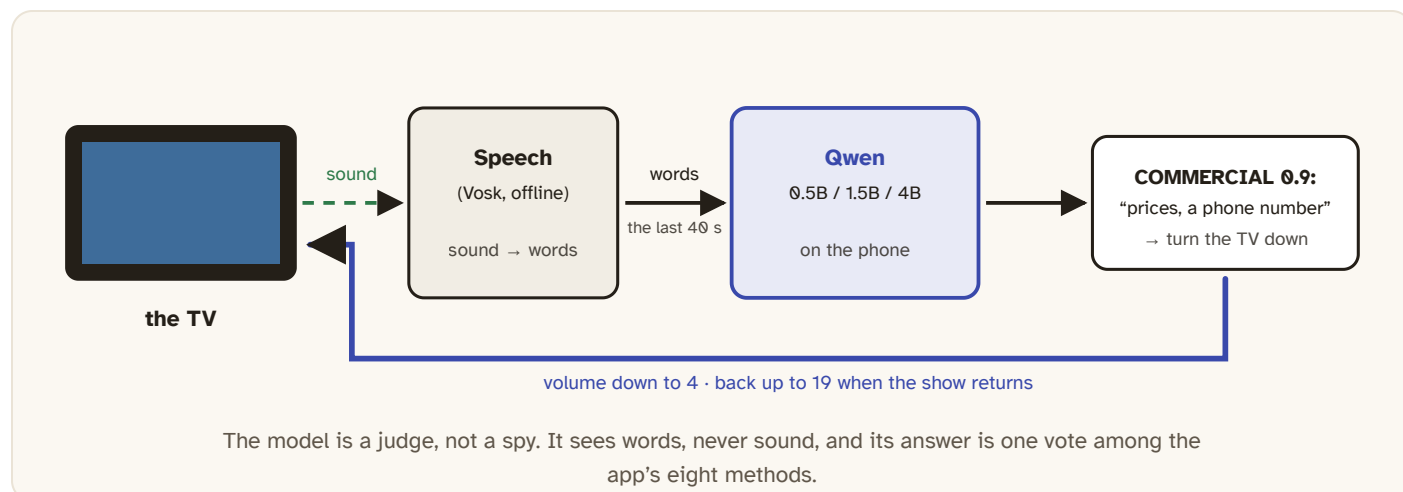
OnePlus Ace 6 Ultra (Dimensity 9500, 12 or 16 GB) · AdHush 0.17 · Qwen 0.5B, 1.5B and 4B

Read this first

The model files are large, from half a gigabyte to nearly three, so download them on **Wi-Fi** with the phone plugged in, and keep the app open until the log says *installed*. An interrupted download resumes from where it stopped when you press the button again. The phone runs warmer while the AI is on, which is normal, and it never uploads anything: the words, the question and the answer all stay inside the phone.

Time: 20 minutes plus the download. **Needs:** the AdHush app already talking to your TV (the TV page shows a green Test), about 3 GB free storage for the biggest size, and the Speech method.

How it works, in one picture



The model does not run continuously. In **tie-breaker** mode it is consulted only when the other methods cannot decide, or while the TV is ducked and the app needs to know whether the show is back; in **always** mode it is consulted every ten seconds regardless. A confident commercial answer is also saved as a *script*, so the next time that ad airs the app recognises the words itself without asking anyone.

The three sizes

“B” means billions of parameters, the adjustable weights inside the model, and more of them means better judgement at the price of a bigger download, more memory and slower answers. All three are Qwen models from Alibaba, converted for phones by the LiteRT community on Hugging Face under the Apache-2.0 licence, which allows this use without an account or a login.

Size	Download	Memory it needs	An answer takes	What it is like
Qwen 2.5 – 0.5B	550 MB	about 1 GB free	1–2 seconds	Quick and cheap. Catches obvious ads. Hedges on tricky ones and is easily fooled by garbled speech.
Qwen 2.5 – 1.5B	1.6 GB	about 2.5 GB free	3–5 seconds	The sweet spot on this phone. Noticeably sharper; still fast enough to catch an ad early.
Qwen 3 – 4B	2.7 GB	about 3 GB free (more on the CPU)	6–12 seconds	The best judgement. Slower, so use it as a tie-breaker. Uses the newer <code>.litertlm</code> file format.

Why 4B and not 3B? No phone-ready Qwen chat model exists at exactly 3 billion parameters: the ready-made bundles jump from 1.5B straight to Qwen 3 at 4B, so that is the “large” choice. Your OnePlus has 12 or 16 GB of memory, which is plenty for it, whereas a phone with 8 GB should stay on 0.5B or 1.5B.

The times are for a Dimensity 9500 that is awake and cool; a hot phone slows down noticeably. Once the file is on the phone none of this needs the internet.

1 Give the AI something to hear

The model judges words, so something has to produce words first, and on this phone that is **Method 5, Speech**: an offline recogniser that turns what the microphone hears into text.

1. Open AdHush and tap **Methods** at the bottom.
2. Find card **5. Speech**. Choose a speech model: *small* (40 MB, quick) or *medium* (128 MB, hears better over a noisy room). With loud fans in the room, choose medium.
3. Tap **Download the speech model** and wait for *installed*.
4. Switch on **Use speech**. If the phone asks for the microphone, say yes.

Captions work too: if the TV shows closed captions and the camera is set up, **Method 6** reads the words off the screen and feeds the same judge, which is cleaner than speech in a noisy room but needs the camera aimed first. Speech is the easier place to start.

2 Pick a size and download it

1. On the Methods page, scroll to card **8. Local AI (on the phone)**.

- 2. Under **Which model**, pick a size. Start with **Qwen 2.5 1.5B** on this phone. You can come back for the others; every size keeps its own file.
- 3. Tap the big button. It says **Download Qwen 2.5 — 1.5 billion (1599 MB)**.
- 4. Watch the **Log** tab. It prints the percentage every few seconds. On home Wi-Fi the 1.5B file takes two to five minutes; the 4B file about twice that.
- 5. When it says *installed — switch on Local AI and Start*, the button changes to **Local AI model: ... installed**.

Lost Wi-Fi halfway? Tap the button again. The app keeps the part it has and continues from there.

3 Switch it on and test it

- 1. On card 8, switch on **Use the local AI**.
- 2. Tap **Test the local AI with a sample**. The app feeds it two made-up transcripts, one from a show and one from an ad, and prints the answers in the Log with the time each took.
- 3. A good answer looks like **COMMERCIAL 0.9: prices and a phone number** for the ad and **SHOW 0.8: a discussion between hosts** for the show. The number is how sure it is, from 0 to 1.

The first answer after a Start is slow because the model is being loaded into memory; later answers are faster. If the test prints *failed to load*, the fix-it table at the end covers the likely causes.

4 Decide how often to ask

Also on the Methods page, under **How often to ask**:

Choice	What happens	Best with
Tie-breaker	Ask only when the other methods are unsure, or while the TV is ducked. Otherwise at most once a minute.	1.5B and 4B. Keeps the phone cool.
Always	Ask every ten seconds, whatever the others think.	0.5B, or when the AI is your main method.

Type your channel’s name in the **Channel** box on card 7 if you have not already, because the judge is told which channel it is listening to, and knowing that it is a news channel helps it tell an anchor reading a headline from a pitchman reading a price.

5 Start, and watch it think

- 1. Go to **Home** and tap **Start**. Put the phone where the microphone can hear the TV.
- 2. The status line shows the app’s state and, when the AI has spoken, its last answer, for example *local: COMMERCIAL 0.85*.
- 3. On the Methods page, card 8’s dot turns the indigo colour while the AI is running. The chips on Home show which methods are on.

4. When the AI is right, do nothing. When it ducks the show by mistake, tap **X Not an ad**: the app brings the sound back and ignores every method for a minute.
5. When it misses an ad, tap **✓ Is an ad** at the start and **► Show's back** at the end. The app learns that break by sound and by its words, so it needs no AI next time.

On the **Log** tab every AI answer appears with its reason, and reading a dozen of them during an evening is the quickest way to judge whether the size you chose is sharp enough for your channel.

6 Switching sizes

Each size is its own file, so switching is simply **Methods** → card 8 → pick another size → download it if the button offers to → **Stop** and **Start** on Home; the previous file stays on the phone. All three files together take about 4.9 GB, which this phone has room for, and nothing else in the app gets slower because of them.

To reclaim the space, Android's *Settings* → *Apps* → *AdHush* → *Storage* → *Clear data* removes the model files, but it also removes the app's settings, learned ads and scripts, so only do that when you want a fresh start.

? Which size should I use?

1. **Start with 1.5B in tie-breaker mode.** On a Dimensity 9500 it answers in a few seconds and is sharp enough for most channels.
2. **Drop to 0.5B** if you want the AI in *always* mode, or if the phone is running hot with the camera on as well.
3. **Move up to 4B** if the log shows the AI hedging (answers near 0.5) or calling the show a commercial. Keep it in tie-breaker mode and keep the phone plugged in.
4. Remember that it is one vote among eight: the fingerprints, the camera bug and the scripts do most of the daily work, and the AI earns its place on the ads none of them has met yet.

Fix-it table

What you see	What to do
The Log says <i>download failed ... press again to resume</i>	Check Wi-Fi and tap the button again. It continues from where it stopped.
Card 8 says <i>no model</i> or the status line says <i>local: no model</i>	The download did not finish, or a different size is selected than the one you downloaded. Pick the size you have, or download the one selected.
<i>local AI failed to load</i> after Start	Not enough free memory: close other apps, or choose a smaller size. With the 4B file this can also mean the app's runtime cannot read the newer format yet; use 1.5B and tell the project.
The AI never answers; the status line has no <i>local:</i> part	It has no words. Check that Speech (or Captions) is on and its model is installed, and that the microphone permission was given.
Answers take much longer than the table says	The phone is hot or on battery saver. Plug it in, give it air, or use tie-breaker mode or a smaller size.
It calls the show a commercial	Tap X Not an ad . If it keeps happening, move up a size, or set the AI to tie-breaker so the other methods must agree.
It misses ads	Teach them with ✓ and ►; try the medium speech model so the words are cleaner; try a bigger size.
The phone is warm	Normal while the AI runs. Tie-breaker mode and a smaller size keep it cooler. The 8,600 mAh battery is big, but plugging in is still better for an evening.

AdHush is free software (MIT). This guide matches AdHush 0.17 on Android. The models are `litert-community/Qwen2.5-0.5B-Instruct`, `litert-community/Qwen2.5-1.5B-Instruct` and `litert-community/Qwen3-4B-Instruct-2507` on Hugging Face, Apache-2.0. The app downloads them straight from there; nothing else is ever sent or fetched. The phone in the title is the OnePlus Ace 6 Ultra; any Android phone with 6 GB or more can run the two smaller sizes.